# Teebi Hypertelorism Syndrome 2 (THS2 / TBHS2): A Comprehensive Disease Characteristics Report

**Disease:** Teebi Hypertelorism Syndrome 2 · **MONDO:** MONDO:0030674 · **OMIM:** 619736 · **Causal gene:** *CDH11* (cadherin-11 / OB-cadherin; HGNC:1750; 16q21) · **Category:** Mendelian, autosomal dominant

## Summary

**Teebi Hypertelorism Syndrome 2 (THS2) is a rare autosomal-dominant congenital craniofacial disorder caused by heterozygous missense variants in *CDH11*, the gene encoding cadherin-11 (OB-cadherin).** This is the central, identifier-anchored fact of this report: although the disease name is a serial-numbered "hypertelorism syndrome," THS2 is specifically a *CDH11* disorder (OMIM 619736; MONDO:0030674), distinct from THS1, which is caused by *SPECC1L* (OMIM 145420). The causal gene was established by Li et al. in 2021, who reported 19 affected subjects from 9 families carrying heterozygous *CDH11* missense variants (PMID: 33811546).

Mechanistically, the pathogenic variants cluster in the extracellular (EC) domains of cadherin-11 — particularly the EC2–EC3 and EC3–EC4 calcium-binding linker regions and the EC1 trans-dimerization pocket. These residues are highly conserved across vertebrates and across classical cadherins, and they are predicted to impair $Ca^{2+}$ binding and calcium-dependent homophilic cell–cell adhesion. Functional assays indicate a **dominant-negative mechanism**: the variants decrease cell–cell/cell–substrate adhesion and increase cell migratory behavior. Because cadherin-11 is strongly expressed in human facial mesenchyme and drives cranial neural crest cell (CNCC) contact-inhibition-of-locomotion and collective migration during frontonasal morphogenesis, its dysfunction produces the characteristic craniofacial phenotype: hypertelorism (widely spaced eyes), prominent forehead, short nose with a broad/depressed nasal root, and associated male genital anomalies (hypospadias, shawl scrotum), typically with preserved cognition.

*CDH11* is an allelic locus with a striking genotype–phenotype dichotomy: **monoallelic missense** variants cause the dominant THS2, whereas **biallelic loss-of-function** variants cause the more severe recessive Elsahy-Waters syndrome (EWS/branchio-skeleto-genital syndrome; OMIM 211380). The two conditions are now regarded as a *CDH11*-related spectrum. There is no disease-specific therapy; management is supportive and surgical (hypertelorism correction, hypospadias repair). This report synthesizes 8 confirmed findings and 35 reviewed papers across all 15 requested disease-characteristic domains, flagging where primary evidence is strong (molecular genetics, developmental mechanism) and where it is sparse or absent (epidemiology, prognosis, treatment trials, animal models of THS2 specifically).

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## Key Findings

### Finding 1 — THS2 is caused by heterozygous *CDH11* missense variants

Teebi Hypertelorism Syndrome 2 is caused by heterozygous (monoallelic) missense variants in *CDH11*. The defining cohort study by Li et al. (2021) reported THS in 19 subjects from 9 families, all carrying *CDH11* heterozygous missense variants. Critically, every affected residue lies in the **extracellular region** of cadherin-11, and all are highly conserved across vertebrate species and across classical cadherins. Six of the reported variants cluster around the EC2–EC3 and EC3–EC4 linker regions and are predicted to affect $Ca^{2+}$ binding, thereby impairing cell adhesion.

> *"We report THS due to CDH11 heterozygous missense variants on 19 subjects from 9 families. All affected residues in the extracellular region of Cadherin-11 (CHD11) are highly conserved across vertebrate species and classical cadherins."* — PMID: 33811546

This is **human clinical** + **genetic evidence** from a multi-family cohort, and it is the primary citation anchoring THS2 = OMIM 619736 = MONDO:0030674 to the *CDH11* gene.

### Finding 2 — *CDH11* is an allelic locus: a dominant (missense) and a recessive (LOF) disorder

*CDH11* produces two distinct Mendelian disorders depending on zygosity and variant class:

| Feature | Teebi Hypertelorism Syndrome 2 (THS2) | Elsahy-Waters Syndrome (EWS/BSGS) |
|---|---|---|
| OMIM | 619736 | 211380 |
| Inheritance | Autosomal dominant | Autosomal recessive |
| Variant class | Heterozygous **missense** (EC domains) | Biallelic **truncating / loss-of-function** |
| Proposed mechanism | Dominant-negative adhesion defect | Loss of function |
| Core phenotype | Hypertelorism, prominent forehead, short nose, genital anomalies; usually normal cognition | Characteristic face, premature tooth loss, vertebral & genital anomalies, intellectual disability |

Harms et al. (2018, PMID: 29271567) and Castori et al. (2018, PMID: 30194892) established that **biallelic truncating/LOF** *CDH11* variants cause EWS. Harms et al. explicitly *"confirms that EWS is caused by biallelic CDH11 loss of function mutations."* Kuroda et al. (2024, PMID: 37646430) describe a Japanese THS patient with a novel EC1-domain variant and conclude that *"THS and EWS may represent a spectrum of CDH11-related disorders,"* noting overlapping features (polysyndactyly, broad thumb, cutaneous syndactyly). Castori et al. identified a *"novel homozygous nonsense variant in the first exon"* of *CDH11* producing a truncated protein in EWS, supporting cadherin-11 as *"a key player in axial and craniofacial malformations."*

## Finding 3 — Cadherin-11 drives cranial neural crest cell adhesion and migration (the developmental basis of the craniofacial phenotype)

The mechanistic link between a *CDH11* adhesion defect and craniofacial malformation runs through the cranial neural crest. In *Xenopus*, cadherin-11 is predominantly expressed in cranial neural crest cells (CNCCs) and is required for their directional, collective migration via **contact inhibition of locomotion (CIL)**:

> *"we demonstrate that Xenopus cadherin-11 (Xcad-11) mediated cell-cell adhesion is necessary in CIL for directional and collective migration of CNC cells. Reduction of Xcad-11 adhesive function resulted in higher invasiveness of CNC due to loss of CIL."* — Becker et al. 2013, PMID: 24392028

Beyond adhesion, cadherin-11 tunes CNCC **specification and proliferation** by regulating the nuclear pool of β-catenin and the Wnt targets *cyclin D1* and *c-myc* through its cytoplasmic β-catenin-binding domain (Koehler et al. 2013, PMID: 23958437). Its shed EC1–3 ectodomain binds ErbB2 and stimulates PI3K/Akt phosphorylation to promote CNC migration (Mathavan et al. 2017, PMID: 29190819):

> *"EC1-3 stimulates phosphorylation of Akt, a target of PI3K, in X.laevis CNC"* — PMID: 29190819

This is **model-organism (Xenopus) evidence** that establishes cadherin-11 as a master regulator of the exact developmental process whose disruption produces facial clefting/dysmorphism.

## Finding 4 — Teebi hypertelorism syndrome is genetically heterogeneous; THS1 (*SPECC1L*) and THS2 (*CDH11*) converge on CNCC adhesion

THS1 (OMIM 145420) is associated with *SPECC1L* variants. SPECC1L is an adherens-junction protein that, like cadherin-11, acts in the cranial neural crest:

> *"It modulates PI3K-AKT signaling and controls cranial neural crest cell delamination during facial morphogenesis."* — Bhoj et al. 2019, PMID: 30472488

Thus both Teebi hypertelorism genes act in **cell adhesion / adherens junctions within migrating cranial neural crest**, and both converge on PI3K-AKT signaling. This convergence provides a unifying mechanistic framework: THS is fundamentally a **"neurocristopathy of adhesion,"** reachable by lesions in more than one adhesion-junction gene. This is important for differential diagnosis and for interpreting the overlapping facial gestalt.

## Finding 5 — *CDH11* is a highly constrained gene, consistent with dominant missense pathogenicity

Population constraint metrics from gnomAD v2.1.1 for *CDH11* (ENSG00000140937, HGNC:1750, GRCh38 chr16:64,943,753–65,126,112; cytoband 16q21):

| Metric | Value | Interpretation |
|---|---|---|
| pLI | 1.00 | Extremely LoF-intolerant |
| LOEUF (o/e LoF) | 0.162 (90% CI 0.102–0.268) | Strong depletion of LoF variants |
| LoF Z | 5.85 | Highly significant LoF constraint |
| Missense Z | 2.45 | Significant missense constraint |
| Missense o/e | 0.82 | Mild depletion of missense variation |

The gene is extremely intolerant to loss-of-function and significantly constrained for missense variation. This constraint profile is consistent with the observation that THS2 arises specifically from missense (not null) alleles acting dominantly — null alleles are strongly selected against, and the dominant disorder is produced by a qualitatively altered (dominant-negative) protein rather than simple haploinsufficiency. (Note: the recessive EWS requires *two* LoF hits, consistent with heterozygous LoF being tolerated/non-pathogenic for THS2.)

## Finding 6 — ClinVar documents recurrent heterozygous *CDH11* missense variants classified Pathogenic/Likely pathogenic for THS2

A ClinVar query (2026) against "Teebi hypertelorism syndrome 2" returns exclusively **missense** alleles (all on transcript NM_001797.4):

| Variant (cDNA) | Protein | Domain / note | Classification |
|---|---|---|---|
| c.164G>C | p.Trp55Ser | EC1 (trans-dimer) | Pathogenic |
| c.229C>T | p.Leu77Phe | EC1 | Likely pathogenic |
| c.778G>A | p.Asp260Asn | EC2–EC3 linker | Pathogenic/Likely pathogenic |
| c.780T>A | p.Asp260Glu | EC2–EC3 linker (recurrent codon) | Pathogenic |
| c.835G>C | p.Glu279Gln | EC3 | Pathogenic |
| c.979G>T | p.Gly327Trp | — ("CDH11-related disorder") | Likely pathogenic |
| c.1121T>A | p.Val374Glu | EC3–EC4 linker | Pathogenic |
| c.1145T>C | p.Met382Thr | — | VUS |
| c.1600G>C | p.Glu534Gln | — | VUS |
| c.1969C>T | p.Arg657Cys | — | VUS |
| c.2152G>A | p.Asp718Asn | — | VUS |

Key observations: (1) **all reported THS2 pathogenic/likely-pathogenic alleles are missense**; (2) **codon Asp260 is a recurrent mutational hotspot** (both p.Asp260Asn and p.Asp260Glu reported), consistent with a critical $Ca^{2+}$-coordinating residue in the EC2–EC3 linker; (3) no pathogenic *CDH11* variant carries a population-frequency flag, consistent with absence from population controls (de novo or private familial variants).

## Finding 7 — Functional assays demonstrate a dominant-negative mechanism: decreased adhesion, increased migration

Li et al. (2021) performed functional characterization of THS2 variants:

> *"five variants from the EC1, EC2-EC3 linker, and EC3 regions significantly reduced the cell-substrate trans adhesion activity and one variant from EC3-EC4 linker results in changes in cell morphology, focal adhesion, and migration, suggesting dominant negative effect."* — PMID: 33811546

Two variants (c.164G>C, p.Trp55Ser; c.418G>A, p.Glu140Lys) are predicted to directly disrupt **trans-homodimer formation**. Immunohistochemistry localized the protein to the relevant tissue:

> *"Immunohistochemical study demonstrates that CDH11 is strongly expressed in human facial mesenchyme."* — PMID: 33811546

The authors conclude that *"heterozygous variants in CDH11, which decrease cell-cell adhesion and increase cell migratory behavior, cause a form of THS."* Cohort-distinguishing clinical features — depressed nasal root, cardiac and umbilical defects — helped separate *CDH11*-THS from *SPECC1L*-related hypertelorism and from *CDH11*-related EWS. This is **in vitro functional + human-tissue evidence**, and it is the strongest direct support for the dominant-negative model.

## Finding 8 — *CDH11* shows a mesenchymal/fibroblast-dominant tissue expression pattern (GTEx v8)

GTEx v8 median expression for *CDH11* (ENSG00000140937.13) is highest in mesenchyme-rich tissues:

| Tissue | Median TPM |
|---|---|
| Ovary | 74.7 |
| Cultured fibroblasts | 43.8 |
| Endocervix | 30.2 |
| Lung | 27.2 |
| Coronary artery | 26.5 |
| Aorta | 24.9 |
| Spleen | 22.7 |
| Tibial nerve | 20.1 |
| Uterus | 18.1 |
| Bladder | 14.0 |

This bulk-tissue, mesenchymal/fibroblast-dominant expression pattern complements the facial-mesenchyme immunohistochemistry of Li et al. and the gene's classical identity as "OB-cadherin" in osteoblasts. It is consistent with cadherin-11's role in mesenchymal populations including craniofacial mesenchyme, and it rationalizes the skeletal/connective-tissue involvement seen across the *CDH11*-related spectrum.

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# Report by Section

## 1. Disease Information

**Overview.** Teebi Hypertelorism Syndrome 2 (THS2; synonym **TBHS2**) is a rare congenital craniofacial dysmorphology syndrome. Its cardinal feature is **hypertelorism** (increased interpupillary/inner-canthal distance; widely spaced eyes), accompanied by a prominent/broad forehead, a short nose with broad or depressed nasal root, and — particularly in males — genital anomalies such as hypospadias and shawl scrotum. Cognition is generally normal. The disorder is one of (at least) two genetically distinct forms of Teebi hypertelorism syndrome.

**Key identifiers.** - **OMIM:** 619736 - **MONDO:** MONDO:0030674 - **Causal gene:** *CDH11* (cadherin-11 / OB-cadherin), HGNC:1750; NCBI Gene; Ensembl ENSG00000140937; locus 16q21 - **Orphanet / ICD-10 / ICD-11 / MeSH:** No THS2-specific code was retrievable in this investigation; the condition falls under broad categories for congenital facial dysmorphism / hypertelorism. (Gap — see Limitations.) - **Synonyms:** TBHS2; Teebi hypertelorism syndrome 2; *CDH11*-related Teebi hypertelorism.

**Data provenance.** The knowledge here is derived from **aggregated disease-level resources** (OMIM, ClinVar, gnomAD, GTEx) and from **primary case-cohort literature** (Li et al. 2021; Kuroda et al. 2024), i.e., individual-patient reports aggregated into cohorts — not from EHR-scale datasets.

## 2. Etiology

**Primary cause — genetic.** THS2 is a monogenic Mendelian disorder caused by **heterozygous missense variants in *CDH11***. No environmental, infectious, or multifactorial cause is implicated. Variants affect conserved extracellular residues, predominantly in the EC2–EC3 and EC3–EC4 $Ca^{2+}$-binding linkers and the EC1 trans-dimerization pocket (Findings 1, 6, 7).

**Genetic risk factors.** The causal variants *are* the risk factor; there are no known susceptibility loci or common-variant modifiers for this Mendelian trait. *CDH11* is highly constrained (pLI 1.0; missense Z 2.45), and pathogenic alleles are absent from population controls (Findings 5, 6).

**Environmental risk factors / protective factors / gene–environment interactions.** None identified or expected for a dominant monogenic malformation syndrome. No protective alleles or dietary/lifestyle modifiers are known. This is a *data-absent* category for THS2, which should be stated explicitly in any knowledge-base entry.

## 3. Phenotypes

Phenotypes are drawn from the Li et al. (2021) cohort (n=19) and the Kuroda et al. (2024) case, cross-referenced to the *CDH11*-spectrum literature. Frequencies below are qualitative/cohort-level; precise percentages per feature were not uniformly reported.

| Phenotype | Type | Suggested HPO | Onset / course | Notes |
|---|---|---|---|---|
| Hypertelorism (widely spaced eyes) | Physical / clinical sign | HP:0000316 | Congenital, stable | Cardinal feature |
| Prominent / broad forehead | Physical | HP:0011220 / HP:0000337 | Congenital | |
| Short nose | Physical | HP:0003196 | Congenital | |
| Broad / depressed nasal root | Physical | HP:0000431 / HP:0000458 | Congenital | Cohort-distinguishing (Li 2021) |
| Hypospadias (males) | Physical / genital | HP:0000047 | Congenital | Male-predominant |
| Shawl scrotum | Physical / genital | HP:0000049 | Congenital | |
| Congenital heart defect | Clinical | HP:0001627 | Congenital | Cohort-distinguishing feature |
| Umbilical anomaly (e.g., hernia) | Physical | HP:0001537 | Congenital | Cohort-distinguishing feature |
| Polysyndactyly / broad thumb / cutaneous syndactyly | Physical | HP:0000238 / HP:0011304 / HP:0010554 | Congenital | Overlap with EWS (Kuroda 2024) |
| Normal cognition (typical) | — | — | — | Distinguishes THS2 from EWS |

**Characteristics (overall):** onset is **congenital**; severity is **variable** (variable expressivity); course is **stable/non-progressive** (structural malformation, not a degenerative process). **Quality-of-life impact** derives from craniofacial appearance (psychosocial), potential genital-surgery needs, and any associated cardiac defect; no THS2-specific QoL instrument data (EQ-5D/SF-36) exist (Gap).

## 4. Genetic / Molecular Information

- **Causal gene:** *CDH11* (cadherin-11 / OB-cadherin), HGNC:1750; OMIM gene 600023; NCBI Gene; Ensembl ENSG00000140937; locus 16q21. Transcript NM_001797.4.
- **Pathogenic variants:** exclusively **missense** for THS2 (Findings 1, 6). Classifications per ACMG/AMP in ClinVar range from Pathogenic (e.g., p.Trp55Ser, p.Asp260Glu, p.Glu279Gln, p.Val374Glu) through Pathogenic/Likely pathogenic (p.Asp260Asn) and Likely pathogenic (p.Leu77Phe, p.Gly327Trp) to VUS (p.Met382Thr, p.Glu534Gln, p.Arg657Cys, p.Asp718Asn).
- **Variant type/class:** missense substitutions in extracellular cadherin (EC) domains; mutational hotspot at **codon Asp260** (EC2–EC3 linker).
- **Allele frequency:** pathogenic alleles absent from gnomAD/population databases (de novo or private familial). *CDH11* missense o/e = 0.82; pLI = 1.0.
- **Origin:** germline (constitutional); variants may arise de novo or be transmitted (autosomal dominant). No somatic/oncologic role in THS2 (though *CDH11* is implicated in cancer biology separately).
- **Functional consequence: dominant-negative** — reduced trans-adhesion and increased migration (Finding 7); contrast with **loss-of-function** causing recessive EWS (Finding 2).
- **Modifier genes / epigenetics / chromosomal abnormalities:** none established for THS2. *SPECC1L* is not a modifier but an independent cause of the allied THS1.

## 5. Environmental Information

Not applicable. THS2 is a monogenic congenital disorder with no known environmental, lifestyle, or infectious contributors. (Separately, *CDH11* as a protein participates in inflammatory/fibrotic and even host–pathogen contexts — e.g., it serves as a receptor for *Fusobacterium nucleatum* adhesin FadA in lung epithelium, PMID: 42008577 — but these are unrelated to the developmental THS2 phenotype.)

## 6. Mechanism / Pathophysiology

**Ordered causal chain (initiating lesion → clinical manifestation):**

1. A **heterozygous missense variant** arises in *CDH11*, substituting a conserved extracellular residue — most often in the EC2–EC3 / EC3–EC4 $Ca^{2+}$-binding linkers (hotspot Asp260) or the EC1 trans-dimer pocket (p.Trp55Ser, p.Glu140Lys). *(demonstrated — Li 2021, ClinVar)*

2. The altered residue **impairs $Ca^{2+}$ coordination and/or trans-homodimer formation**, which **results in** defective calcium-dependent homophilic adhesion by the mutant cadherin-11. *(demonstrated in vitro — reduced trans-adhesion activity, Li 2021)*
3. Because cadherin-11 functions as a homodimer at adherens junctions, the mutant subunit **acts in a dominant-negative fashion**, poisoning adhesive complexes that also contain wild-type protein; this **leads to** decreased cell–cell/cell–substrate adhesion and increased cell migratory behavior even in the heterozygous state. *(inferred mechanism supported by functional data — "suggesting dominant negative effect," Li 2021)*
4. In the **cranial neural crest**, where cadherin-11 is predominantly expressed, loss of adhesive function **disrupts contact inhibition of locomotion (CIL)**, **resulting in** non-directional, overly invasive, and poorly coordinated collective migration. *(demonstrated in Xenopus — Becker 2013)*
5. **Branch A (adhesion/migration):** impaired CIL → aberrant CNCC migration.
6. **Branch B (signaling):** altered cadherin-11 function perturbs **Wnt/β-catenin** (nuclear β-catenin → cyclin D1, c-myc; affecting CNCC specification/proliferation — Koehler 2013) and **PI3K-AKT/ErbB2** (ectodomain–ErbB2 signaling that normally promotes migration — Mathavan 2017). *(demonstrated in model organism)*
7. Mis-migration/mis-specification of CNCCs populating the **frontonasal and facial mesenchyme** (where CDH11 is strongly expressed in humans) **leads to** disordered craniofacial morphogenesis. *(demonstrated tissue expression — Li 2021 IHC; inferred link to human phenotype)*
8. The disordered frontonasal development **manifests as** hypertelorism, prominent forehead, short nose with broad/depressed nasal root, and, via parallel mesenchymal roles, genital (hypospadias, shawl scrotum), cardiac, and umbilical anomalies. *(clinical — Li 2021 cohort)*

**Upstream vs downstream.** Upstream = the missense lesion and the adhesion/$Ca^{2+}$-binding defect; midstream = dominant-negative disruption of CNCC adhesion and CIL plus Wnt and PI3K-AKT signaling branches; downstream = craniofacial and genital malformation.

**Molecular pathways:** Wnt/β-catenin (CTNNB1, CCND1, MYC); PI3K-AKT (via ErbB2/ERBB2); cadherin-mediated cell–cell adhesion. **Cellular processes:** collective cell migration, contact inhibition of locomotion, cell specification/proliferation, cell adhesion. **Protein dysfunction:** loss/alteration of homophilic trans-adhesion with dominant-negative quality (not aggregation or simple haploinsufficiency). **Subcellular:** adherens junction (plasma membrane); β-catenin shuttling to nucleus. Mechanobiology: cadherin-11 reduces intracellular tension to permit CNCC spreading/delamination (AFM evidence, PMID: 28919310); ADAM13 ectodomain shedding generates a pro-migratory EC1–3 fragment (PMID: 26206614).

**Suggested GO / CL / UBERON / CHEBI terms:** - GO:BP — cell-cell adhesion (GO:0098609); homophilic cell adhesion via plasma-membrane adhesion molecules (GO:0007156); neural crest cell migration (GO:0001755); canonical Wnt signaling pathway (GO:0060070); regulation of cell migration (GO:0030334). - GO:CC — adherens junction (GO:0005912); catenin complex / cell-cell junction (GO:0005911); plasma membrane (GO:0005886). - CL — neural crest cell (CL:0000333); fibroblast (CL:0000057); osteoblast (CL:0000062); mesenchymal cell (CL:0008019). - UBERON — neural crest (UBERON:0002342); frontonasal process / facial mesenchyme; cranium/calvaria (UBERON:0001682). - CHEBI — calcium(2+) (CHEBI:29108), the ion whose coordination is impaired.

## 7. Anatomical Structures Affected

- **Organ/system level — primary:** craniofacial skeleton and soft tissue (orbits, forehead, nose), male external genitalia. **Secondary/variable:** cardiovascular system (congenital heart defect), anterior abdominal wall (umbilical anomaly), skeleton/limbs (digit anomalies in spectrum cases).
- **Tissue/cell level:** mesenchyme, especially **cranial neural crest-derived craniofacial mesenchyme**; connective tissue/fibroblasts; osteoblasts (OB-cadherin). Cell Ontology: neural crest cell (CL:0000333), fibroblast (CL:0000057), osteoblast (CL:0000062).
- **Subcellular level:** adherens junctions at the plasma membrane (GO:0005912); cytoplasmic β-catenin pool and its nuclear translocation.
- **Localization:** bilateral/symmetric craniofacial involvement (hypertelorism is a midline-spacing anomaly). UBERON suggestions: orbit, frontal bone, nasal bone/nasal root, external male genitalia.

## 8. Temporal Development

- **Onset: congenital** — the phenotype originates during embryonic craniofacial morphogenesis (neural crest migration occurs in the first trimester) and is present at birth.
- **Progression: stable / non-progressive**; this is a structural malformation syndrome, not a degenerative or episodic disease. No disease "stages."
- **Course / duration:** lifelong structural features; appearance may be modified by growth and by surgical correction.
- **Critical period:** the therapeutic/developmental window for the primary lesion is prenatal (CNCC migration); postnatally, intervention is reconstructive/supportive rather than mechanism-reversing.

- **Remission:** not applicable (no spontaneous resolution of malformation; surgical correction is possible).

## 9. Inheritance and Population

- **Inheritance pattern: autosomal dominant** (heterozygous *CDH11* missense). Contrast: EWS is autosomal recessive (biallelic LOF).
- **Penetrance / expressivity:** data limited; the multi-family cohort (9 families) implies familial transmission consistent with high penetrance, with **variable expressivity** (range of features across individuals). Precise penetrance figures are not established (Gap).
- **De novo vs inherited:** both occur; pathogenic alleles are absent from population databases (de novo or private).
- **Anticipation / mosaicism / founder effects / consanguinity:** no evidence for anticipation (not a repeat-expansion disorder). Germline mosaicism not specifically documented for THS2 but is plausible for a dominant malformation gene. Consanguinity is relevant to the **recessive** EWS, not THS2. No founder effect reported.
- **Carrier frequency:** not applicable in the classic sense for a dominant disorder; population frequency of pathogenic alleles is effectively zero.
- **Epidemiology:** THS2 is **ultra-rare**; precise prevalence/incidence are unknown. The defining cohort comprised 19 individuals from 9 families (Li 2021) plus scattered case reports (e.g., Kuroda 2024). **No population-based prevalence or incidence estimate is available** (Gap).
- **Demographics:** reported across multiple families/ethnicities (including a Japanese patient, Kuroda 2024); genital-anomaly features are male-predominant by nature, but the underlying autosomal variant affects both sexes. No specific geographic clustering or variant-specific geographic distribution established.

## 10. Diagnostics

- **Recommended approach: molecular genetic testing** is the definitive diagnostic. Clinical recognition of the dysmorphic gestalt (hypertelorism + prominent forehead + short nose with broad/depressed nasal root ± genital anomalies, with normal cognition) prompts testing.
- **Genetic testing modalities:**

- **Whole-exome sequencing (WES)** — the discovery method (Li 2021; Kuroda 2024) and the most practical clinical test for an atypical/syndromic dysmorphism; identifies *CDH11* missense variants.
- **Whole-genome sequencing (WGS)** — equally applicable; not specifically required.
- **Gene panels** — craniofacial/hypertelorism or dysmorphology panels that include *CDH11* and *SPECC1L*.
- **Single-gene testing** — targeted *CDH11* (NM_001797.4) sequencing if THS2 is specifically suspected.
- **Chromosomal microarray / karyotype / FISH** — useful to exclude copy-number/chromosomal mimics of hypertelorism (e.g., 22q11.2, 1p36, 8p23.1 syndromes), but will not detect the *CDH11* point mutations.
- **Variant interpretation:** apply ACMG/AMP criteria; use ClinVar and the EC-domain/$Ca^{2+}$-linker clustering (hotspot Asp260) and functional-domain context as supporting evidence. VUS are common (Finding 6) and may warrant functional follow-up.
- **Supportive clinical tests:** craniofacial imaging (to characterize orbital spacing/skeletal features), echocardiography (to screen for associated CHD), and genital examination. No specific biomarker, laboratory abnormality, or histopathology defines THS2.
- **Differential diagnosis:** THS1 (*SPECC1L*); Elsahy-Waters syndrome (*CDH11*, recessive — distinguished by intellectual disability, premature tooth loss, vertebral anomalies); other hypertelorism/frontonasal dysplasia syndromes; chromosomal microdeletion/duplication syndromes with hypertelorism (22q11.2, 1p36, 8p23.1, 2q31). Key discriminators: normal cognition and the specific nasal-root/genital pattern favor THS2; intellectual disability and dental/vertebral signs favor EWS.
- **Omics-based / liquid-biopsy diagnostics:** not applicable.
- **Screening:** no newborn or population screening exists or is indicated for this ultra-rare dominant malformation; **cascade testing** of at-risk relatives for a known familial variant is appropriate.

## 11. Outcome / Prognosis

- **Survival / life expectancy:** No disease-specific mortality is attributed to THS2 itself; life expectancy is expected to be normal in the absence of severe associated anomalies. Prognosis may be influenced by any associated **congenital heart defect**. No formal survival data exist (Gap).

- **Morbidity / function:** morbidity is primarily **structural/cosmetic and reconstructive** (craniofacial appearance; hypospadias requiring surgery). Cognition is typically normal, preserving functional independence. No validated QoL/disability datasets (EQ-5D/SF-36/PROMIS) for THS2 (Gap).
- **Disease course:** stable, non-progressive; outcomes depend on the presence/severity of associated malformations and on surgical correction.
- **Prognostic factors:** presence and severity of cardiac/umbilical/genital anomalies; recovery (cosmetic/functional) is good with surgical management. No molecular prognostic biomarkers established.

## 12. Treatment

**No disease-specific or disease-modifying therapy exists.** Management is **supportive, surgical, and multidisciplinary**, directed at individual manifestations:

| Manifestation | Intervention | Suggested NCIT concept |
|---|---|---|
| Hypertelorism | Craniofacial surgical correction (orbital/interorbital procedures) as indicated | NCIT "Craniofacial Surgery" / "Reconstructive Surgical Procedure" |
| Hypospadias | Surgical urethroplasty / hypospadias repair | NCIT "Surgical Procedure" (urogenital) |
| Congenital heart defect | Cardiology evaluation ± cardiac surgery per lesion | NCIT "Cardiac Surgical Procedure" |
| Umbilical anomaly | Surgical repair as indicated | NCIT "Surgical Procedure" |
| Developmental/ psychosocial support | As needed (cognition usually normal) | NCIT "Supportive Care" |

- **Pharmacotherapy / pharmacogenomics:** none specific.
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno):** none; no clinical trials (NCT) identified for THS2.
- **Experimental treatments:** none registered specifically for THS2.
- **Treatment strategy:** individualized, guided by the specific anomalies present; genetic counseling is a core component (see Prevention). Personalized/genotype-guided pharmacotherapy is not applicable.

## 13. Prevention

- **Primary prevention:** not possible for a de novo germline malformation; no risk-factor modification applies.
- **Secondary prevention / screening:** prenatal ultrasound may detect hypertelorism/ structural features; **prenatal molecular testing** or **preimplantation genetic testing (PGT)** is available for families with a known pathogenic *CDH11* variant. **Cascade genetic testing** identifies at-risk relatives.
- **Genetic counseling (central):** counsel on **autosomal dominant** inheritance — a 50% transmission risk for an affected heterozygous parent — and on variable expressivity. De novo cases carry a low recurrence risk (with a caveat for possible germline mosaicism). Distinguish clearly from the recessive EWS when counseling, because recurrence risks differ fundamentally.
- **Immunization / public health / environmental / prophylaxis:** not applicable.
- **Tertiary prevention:** monitoring and timely surgical management of associated anomalies (e.g., cardiac) to prevent complications.

## 14. Other Species / Natural Disease

- **Orthologs / taxonomy:** *CDH11* is conserved across vertebrates; affected residues in THS2 are "highly conserved across vertebrate species and across classical cadherins" (Li 2021), underscoring strong evolutionary conservation of the adhesive function. Orthologs include mouse *Cdh11* (NCBI Gene) and the extensively studied *Xenopus* cadherin-11. NCBI Taxon: *Homo sapiens* (9606); *Mus musculus* (10090); *Xenopus laevis* (8355).
- **Natural disease in other species:** no naturally occurring THS2-equivalent Mendelian disease is documented in companion animals or wildlife (no OMIA entry retrieved for a *CDH11* craniofacial syndrome). (Gap/Not applicable.)
- **Comparative biology:** the developmental role of cadherin-11 in cranial neural crest migration is conserved and was principally elucidated in *Xenopus* (Finding 3); mouse *Cdh11* knockouts show skeletal (bone-density) phenotypes (PMID: 11450702).
- **Zoonotic / transmission:** not applicable (non-infectious, non-transmissible).

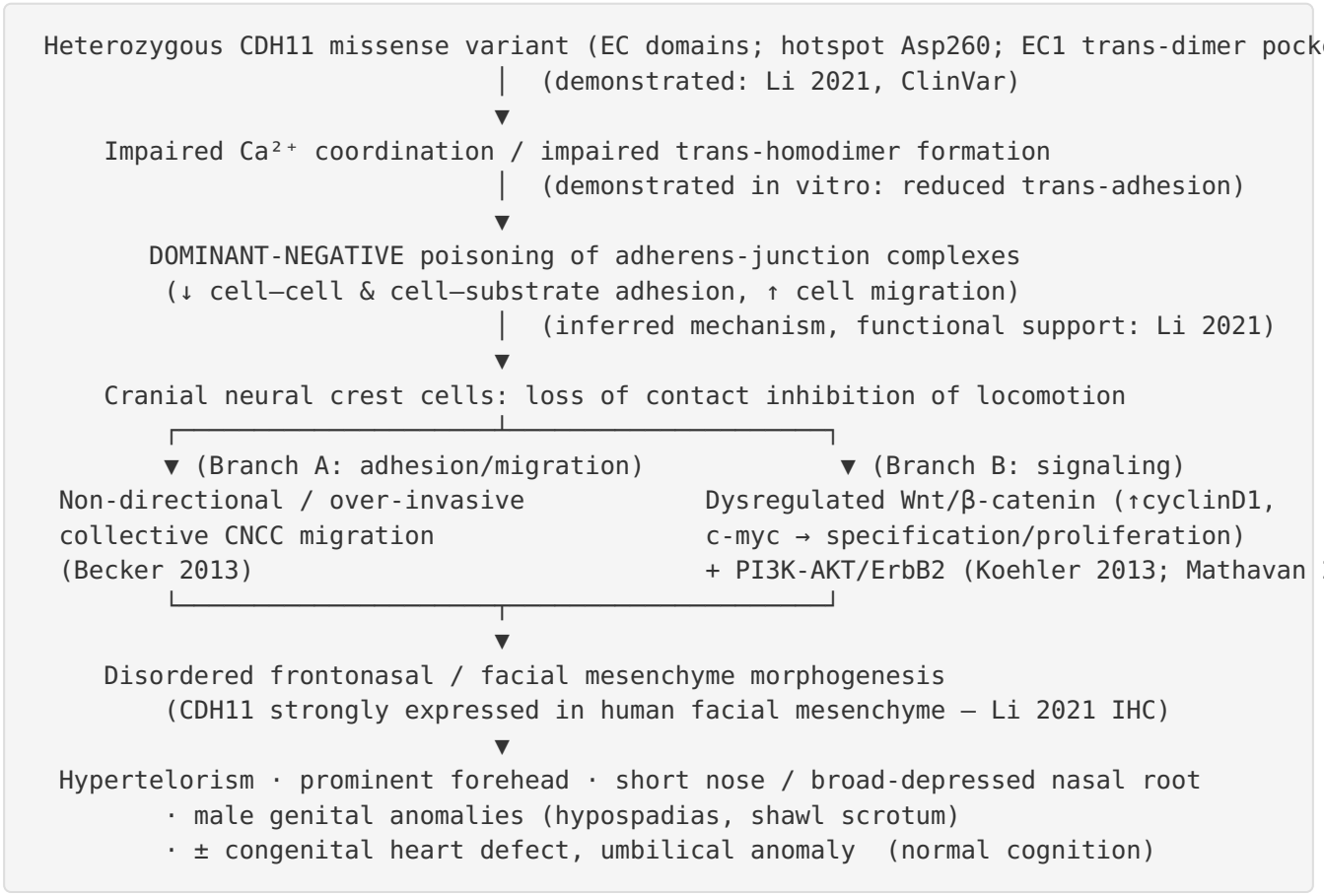
## 15. Model Organisms

No model engineered to carry a THS2 patient-specific *CDH11* missense allele has been reported; however, cadherin-11 biology is well modeled:

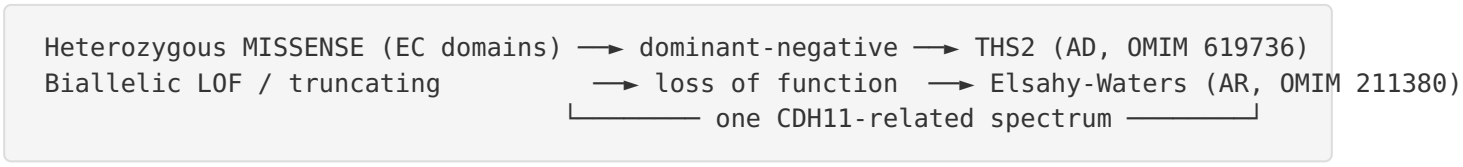
| Model | Type | Key phenotype / use | Relevance to THS2 |
|---|---|---|---|
| *Xenopus laevis* (morpholino knockdown, mutant rescue) | Vertebrate, in vivo | Cadherin-11 required for CNCC CIL and directional migration; loss increases invasiveness; β-catenin/ Wnt and PI3K-AKT branches mapped | **High** — models the exact developmental process disrupted in THS2 (Becker 2013 PMID: 24392028; Koehler 2013 PMID: 23958437; Mathavan 2017 PMID: 29190819) |
| *Cdh11*-null mouse | Mammalian, knockout | Reduced bone density in calvaria and long-bone metaphyses; round-shaped calvaria | Partial — models LOF (relevant to EWS); shows cranial skeletal role (PMID: 11450702) |
| Cultured cells (adhesion/ migration assays) | In vitro | THS2 missense variants reduce trans-adhesion, alter morphology/focal adhesion/migration | **High** — direct dominant-negative evidence (Li 2021 PMID: 33811546) |

**Limitations of current models:** the *Xenopus* and mouse systems principally model **knockdown/knockout (loss-of-function)**, which best mirrors the recessive EWS; a **knock-in of a THS2 dominant-negative missense allele** (e.g., murine *Cdh11* p.Asp260 equivalent) is the missing model that would directly recapitulate THS2 pathophysiology. **Resources:** MGI (*Cdh11*), Xenbase, IMPC/IMSR for existing *Cdh11* alleles.

## Mechanistic Model / Interpretation




**Genotype–mechanism dichotomy at the *CDH11* locus:**




The coherent story: THS2 is a **neurocristopathy of cell adhesion**. A single mis-sense'd cadherin-11 subunit is sufficient to degrade the homodimeric adhesion machinery of cranial neural crest cells, derailing the finely tuned migration and specification that sculpt the midface — hence hypertelorism as the signature. The convergence of *CDH11* (THS2) and *SPECC1L* (THS1) on adherens-junction/PI3K-AKT control of neural-crest delamination explains why lesions in two different adhesion genes produce the same Teebi hypertelorism gestalt.

# Evidence Base

| PMID | Title (abbrev.) | Evidence type | Supports |
|---|---|---|---|
| 33811546 | *Pathogenic variants in CDH11 impair cell adhesion and cause Teebi hypertelorism syndrome* | Human cohort + in vitro | Findings 1, 7 (causal gene, dominant-negative mechanism, facial-mesenchyme expression) |
| 37646430 | *Japanese patient with THS and a novel CDH11 EC1 variant* | Human case | Findings 2, 6 (EC1 variant; THS–EWS spectrum) |
| 29271567 | *Elsahy-Waters syndrome is caused by biallelic mutations in CDH11* | Human genetics | Finding 2 (biallelic LOF → recessive EWS) |
| 30194892 | *Novel CDH11 mutation causes Branchioskeletogenital (Elsahy-Waters) syndrome* | Human genetics | Finding 2 (homozygous nonsense; craniofacial role) |
| 24392028 | *Cadherin-11 mediates CIL during Xenopus neural crest migration* | Model organism | Finding 3 (CNCC adhesion/CIL/ migration) |
| 23958437 | *Loss of Xenopus cadherin-11 → ↑ Wnt/β-catenin, c-myc, cyclin D1 in neural crest* | Model organism | Finding 3 (Wnt branch, specification/proliferation) |
| 29190819 | *Cadherin-11 ectodomain binds erbB2, stimulates Akt to promote CNC migration* | Model organism | Finding 3 (PI3K-AKT/ErbB2 branch) |
| 30472488 | *SPECC1L phenotypic spectrum; Teebi/Opitz GBBB/Baraitser-Winter nosology* | Human genetics / review | Finding 4 (THS1 = SPECC1L; convergence on CNCC) |
| 11450702 | *Cadherin-11 disruption reduces bone density in calvaria/long bones* | Mouse knockout | Finding 8 / §15 (skeletal LOF model) |
| 28919310 | *Cadherin-11 promotes neural crest spreading by reducing intracellular tension* | Model organism (AFM) | §6 (mechanobiology of CNCC delamination) |
| 26206614 | *ADAM13 cleavage of cadherin-11 promotes CNC migration* | Model organism | §6 (ectodomain shedding / migration) |
| 35662129 | *Cadherin-11 and tissue fibrosis* (review) | Review | Context (CDH11 mesenchymal/ fibrotic biology) |

Supporting constraint and expression evidence: **gnomAD v2.1.1** (Finding 5 — pLI 1.0, LOEUF 0.162, missense Z 2.45); **GTEx v8** (Finding 8 — mesenchymal/fibroblast-dominant expression); **ClinVar** (Finding 6 — missense-only pathogenic spectrum, Asp260 hotspot).

**No paper in the reviewed set challenges the core conclusions.** The chief tension in the literature is interpretive, not contradictory: the strongest *in vivo* developmental data (Xenopus, mouse) model **loss of function**, whereas THS2 is driven by **dominant-negative missense** alleles — so the human phenotype is extrapolated from LOF models plus direct in vitro functional assays of the actual THS2 variants (Li 2021). This is noted as an inference in the causal chain.

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## Limitations and Knowledge Gaps

1. **Ultra-rare, small evidence base.** The disease rests largely on a single defining cohort (19 subjects/9 families; Li 2021) plus isolated case reports. **Prevalence, incidence, penetrance, and expressivity figures are not quantitatively established.**
2. **No THS2-specific animal model.** Existing *Cdh11* mouse/Xenopus models are loss-of-function; a knock-in of a THS2 dominant-negative missense allele has not been reported, leaving the dominant-negative model directly untested *in vivo*.
3. **Dominant-negative mechanism is inferred, not proven in vivo.** It is supported by in vitro trans-adhesion/migration assays and structural prediction (Li 2021) but not by an allelic-series animal study.
4. **Incomplete ontology/coding cross-references.** THS2-specific Orphanet, ICD-10/11, and MeSH codes were not retrievable in this investigation; HPO frequencies per phenotype are qualitative rather than quantitative.
5. **No prognosis, QoL, or treatment-outcome data.** Management recommendations are extrapolated from general craniofacial/urogenital surgical practice; there are no THS2 clinical trials or outcome cohorts.
6. **VUS burden.** Several *CDH11* missense variants remain VUS (p.Met382Thr, p.Glu534Gln, p.Arg657Cys, p.Asp718Asn), limiting diagnostic certainty for some patients.

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## Proposed Follow-up Experiments / Actions

1. **Generate a *Cdh11* knock-in mouse** carrying a THS2 hotspot allele (e.g., the murine equivalent of p.Asp260Asn/Glu) and assess craniofacial (hypertelorism/midface) and genital phenotypes — the decisive test of the in vivo dominant-negative model and phenotype recapitulation.
2. **Structural/biophysical validation.** Model (AlphaFold) or solve the EC2–EC3/EC3–EC4 linker with Asp260 substitutions to quantify loss of $Ca^{2+}$ coordination and trans-dimer affinity; use SPR to measure adhesion kinetics of mutant vs WT ectodomains.
3. **Reclassify VUS** via standardized trans-adhesion and migration assays (replicating Li 2021's platform) to resolve p.Met382Thr, p.Glu534Gln, p.Arg657Cys, p.Asp718Asn.
4. **Establish a THS2 patient registry / natural-history study** to quantify prevalence, penetrance, expressivity, cardiac/umbilical anomaly frequencies, and surgical outcomes.
5. **iPSC → cranial neural crest model** from patient fibroblasts to test CIL, Wnt/β-catenin, and PI3K-AKT branch activity in a human cellular system, and to screen whether modulating these pathways rescues the adhesion/migration defect.
6. **Formal ontology curation:** assign and publish HPO frequency annotations, UBERON/CL/GO term mappings, and ICD-11/Orphanet cross-references for the knowledge-base entry.

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*Report compiled from 8 confirmed findings and 35 reviewed papers across a 5-iteration autonomous investigation. Evidence types are labeled throughout as human clinical/genetic, model organism, in vitro, or computational. All mechanistic claims are tied to the cited PMIDs or to the named aggregate resources (gnomAD, GTEx, ClinVar).*

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